5.5 Exponential and Logarithmic Models

Name _____ Period ____ Date _____

Pre-Calculus 5.5 · Student sheet

I can…

- use an exponential growth or decay model to answer questions.
- find an exponential model from two data points.
- solve half-life problems.

Vocabulary · fill in the missing word

Exponential growth model — $y =$ _____, with $b > 0$

Exponential decay model — $y =$ _____, with $b > 0$

Half-life — the time it takes for half of a _____ to decay

Part 1 · Penny decay

Start with 40 pennies. Each round, remove the heads and record what is left.

Round	0	1	2	3	4	5	6
Pennies left	40						

Part 2 · Notes and examples (fill in each note, then work the example)

Growth and decay: $y = ae^{bx}$ grows when b _____ 0 and decays when $b < 0$.

Example 1

A town's population is P = 2.5e^(0.012t) million, where t = 0 is 2010. In what year will it reach 3 million?

Finding a model: use the starting value for a, then substitute a second point and solve for b with _____.

Example 2

Find an exponential growth model $y = ae^{bt}$ through (0, 100) and (5, 300).

Half-life h: in $y = ae^{bt}$, b = _____.

Example 3

Carbon-14 has a half-life of 5730 years. How much of a 10-gram sample remains after 2000 years? Round to the nearest thousandth.

Example 4

A 50-gram sample decays to 40 grams in 3 days. Find its half-life.

Part 3 · Partner practice

Partner 1

A bacteria count is y = 500e^(0.2t) after t hours. When will there be 2000 bacteria?

Partner 2

A medicine has a half-life of 10 days. How much of an 80-mg dose remains after 25 days? Round to the nearest hundredth.